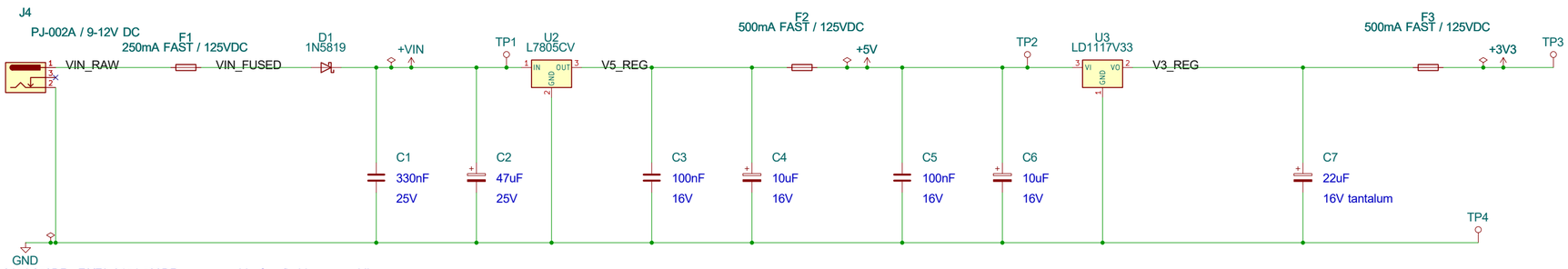
WEATHER GADGET — LCD DISPLAY SUBSYSTEM

Zander Boward | Team 202 | Revision F | 20 October 2026 | Original selected components

ONE A2 SHEET | 3.3 V logic | Three UART links proposed | Button selects the displayed sensor

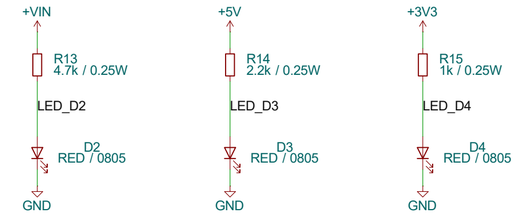
9–12 V INPUT / LINEAR REGULATORS



F1 0154.250DR; F2/F3 0154.500DR — removable fuse/holder assemblies.

U2 tab = GND; U3 tab = OUT. Do not join tabs. C7 stability requires a bench test.

POWER INDICATORS



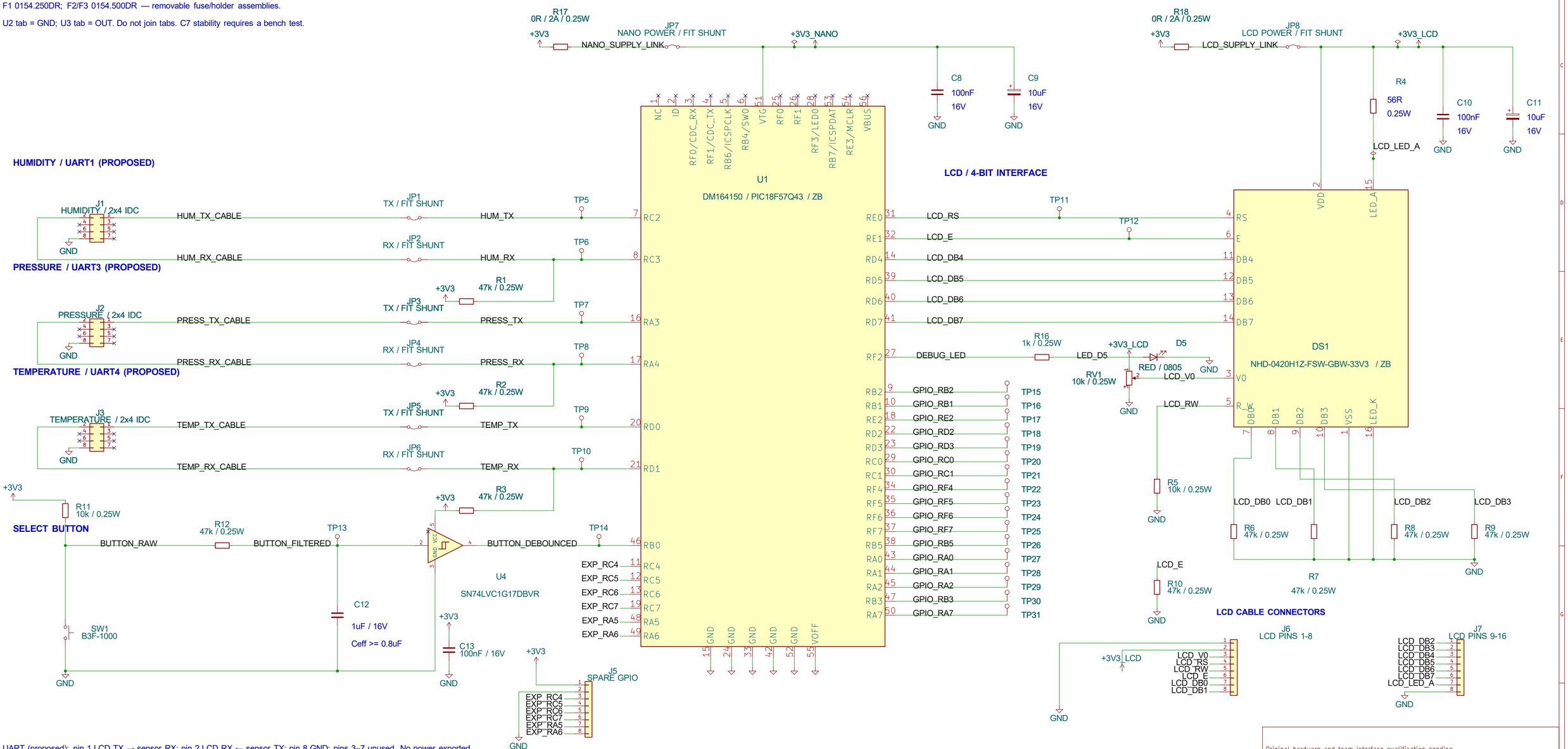
HUMIDITY / UART1 (PROPOSED)

PRESSURE / UART3 (PROPOSED)

TEMPERATURE / UART4 (PROPOSED)

SELECT BUTTON

LCD / 4-BIT INTERFACE



UART (proposed): pin 1 LCD TX → sensor RX; pin 2 LCD RX ← sensor TX; pin 8 GND; pins 3–7 unused. No power exported.

Confirm peer signal voltages and cable continuity. VTG 51 = 3.3 V; VOFF 55 = GND; VBUS 56 unused.

Fit shunts. Power OFF before moving them; signal pins can backfeed unpowered boards. Hardware qualification remains pending.

50 mA Nano allocation is provisional. Verify loaded rails, regulator temperature, C7 stability and backlight brightness.

Original hardware and team interface qualification pending
Revision E circuit retained: readability layout only
Team 202 | Zander Boward
Sheet: /
File: LCD-Board-Connected.kicad_sch
Title: Weather Gadget — LCD Display Subsystem
Size: A2 | Date: 2026-10-20 | **Rev: F**
KiCad E.D.A. 10.0.7 | Id: 1/1